

Can we determine the average speed of gas molecules in a room?

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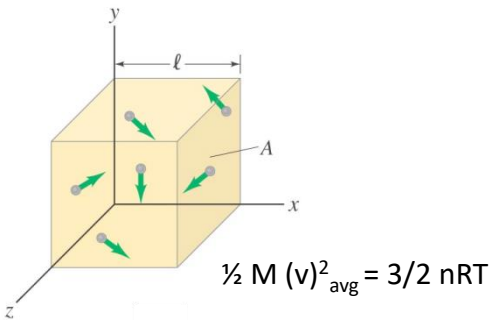
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Week 12

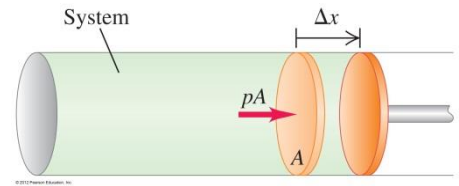
Serway Chap 20.1 – 20.3

"Who sees the future? Let us have free scope for all directions of research; away with dogmatism, either atomistic or anti-atomistic!"

- Ludwig Eduard Boltzmann (1844 - 1906)



<https://www.youtube.com/watch?v=YSTRa27a3BQ&t=5s>



$$W = F\Delta x = P A \Delta x = P \Delta V$$

Self-check – after completing the lessons (lectures and tutorials) for this topic, are you able to

1. use the kinetic theory of gases to determine the root-mean-square speed of a system of gases
2. determine the internal energy and work done by a system of ideal (mono-atomic and diatomic) gases
3. use the first law of thermodynamics to analyze heat engines (using a pV diagram)

Mechanical Energy -> Thermal Energy

In the previous lectures, we have been dealing with macroscopic systems and objects: the forces acting on them; their momenta and the energies associated with them. We have brief encounters of energy lost from macroscopic objects due to friction (and other non-conservative forces) and eventually transformed into heat, sound or light energy. In these cases, essentially we have macroscopic energy (which is highly ordered – molecules in object move together as a single entity with a K.E. or P.E. associated with the object) transforming to microscopic energy (increased vibration and momentum for the individual molecules). At the microscopic level, the movement of the atoms / molecules is highly disordered but is collectively constrained by the conservation of energy and momentum.

In this next two/three lectures, we are going to learn about how the average microscopic energy of the entities that constitute the system is quantified by its TEMPERATURE. The total microscopic energy of a system is collectively quantified as its THERMAL ENERGY (INTERNAL ENERGY) and for different systems in thermal contact, we will study the HEAT transfer between them. In the study of thermodynamics, we are interested in how to extract useful WORK from HEAT and INTERNAL ENERGY.

Outline of lectures:

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Microscopic and
macroscopic properties
of ideal gases

1st law of thermodynamics

Extracting work
from heat

Connecting the Microscopic to Macroscopic world

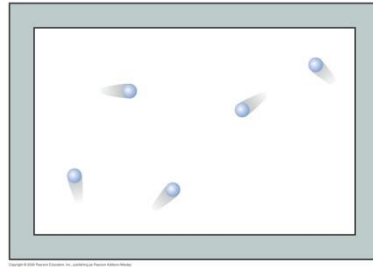
Gas molecules are in constant random motion. While gas molecules are very small and invisible, it is possible to make deductions about their properties by simple principles that we have learnt.

The idea goes this way:

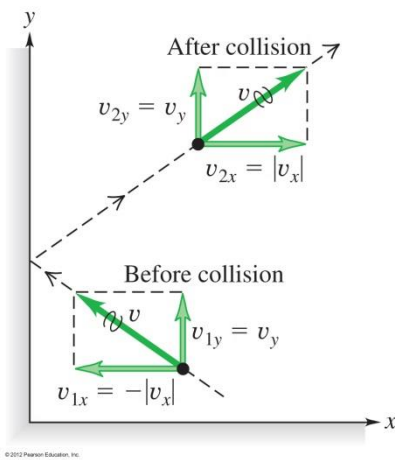
- 1) We can make deductions from Newton's laws about momentum of the gas molecules and the forces they exert on the vessel that contains them.
- 2) Knowing forces, we can compute the pressure $P = F/A$. We note here that the gas pressure is a measurable macroscopic physical quantity and is in the ideal gas equation $PV = nRT$.
- 3) With this connection, we can make inferences about microscopic properties of gases from macroscopic measurements.

Next we will build a theory based on these ideas.

Kinetic Theory of Gases 1



A schematic sketch of gas molecules in container.
[Taken from Knight Chap 16]



[Young(College) pg 487, fig 15.10]

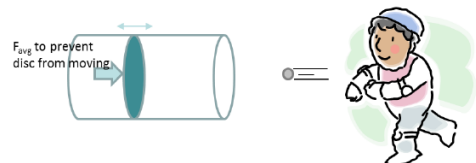
Elastic collision of a molecule of mass m in an idealized container. The component v_y parallel to the wall does not change; the component v_x perpendicular to the wall reverses direction. The speed $|v|$ does not change.

Momentum change per collision
 $= 2m|v_x|$

To compute the average force on the wall, we have to estimate the number of collisions (with wall) per second.

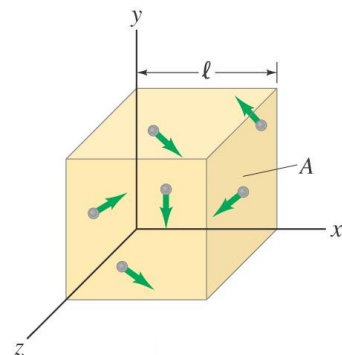
Example:

Throwing 0.02 kg iron balls at 2 m/s horizontally towards at a circular disc which is free to slide (frictionless) within a hollow cylinder placed horizontally. Iron balls bounce back at the same **horizontal** speed in the opposite direction, i.e. -2m/s. If he throws 100 such iron balls in a minute, what is the average horizontal force F_{avg} required to prevent the disc from sliding?

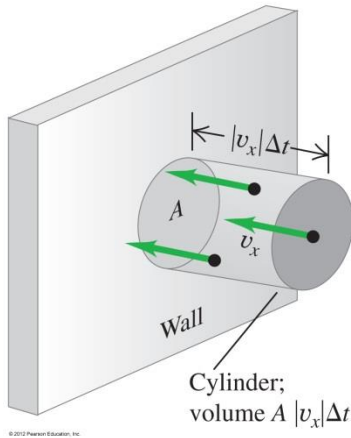


Each hit causes a momentum change of $0.02 \times (2 - (-2)) = 0.08 \text{ kg m/s}$

Average force exerting on disc $F = (100 / 60) \times 0.08 = 0.13 \text{ N}$



Kinetic Theory of Gases 2



[Text pg 487, fig 15.11]
Number of molecules
per unit time colliding
with the wall.

If there are N molecules in the vessel
of volume V , the number of molecules
per unit volume is N/V .

For a given cross sectional area A ,
the number of gas molecule colliding
with wall per second is $\frac{1}{2} A v_x \times 1s \times (N/V)$.
Half of the molecules are moving towards
the wall and half are moving away.
The total change in momentum per s for
the gas molecules in an area A is

$$F = \frac{1}{2} A v_x \times 1s \times (N/V) \times (2m |v_x|) \\ = m |v_x|^2 A (N/V)$$

Thus, the pressure on the wall is
 $p = F/A = m |v_x|^2 (N/V)$.

However, in the container, the gas molecules move freely equally in all three
directions, x-y-z, so $(v_x)^2_{avg} = (v_y)^2_{avg} = (v_z)^2_{avg}$. To compute the magnitude of
the velocity, we note that $(v)^2_{avg} = (v_x)^2_{avg} + (v_y)^2_{avg} + (v_z)^2_{avg}$.

Simplifying, we can write $(v)^2_{avg} = 3(v_x)^2_{avg} \Rightarrow (v_x)^2_{avg} = 1/3 (v)^2_{avg}$.
Therefore,

$$p = F/A = 1/3 m (v)^2_{avg} (N/V).$$

We have made the connection between the microscopic and macroscopic
world. Using the ideal gas equation, $PV = nRT$, we can get other different
forms

$$pV = 1/3 (Nm) (v)^2_{avg} = nRT.$$

The quantity $Nm = M$, the total mass of all gas molecules. We also note that
 $N = n N_A$ where N_A is Avogadro's number. The total (translational) kinetic
energy of all the gas molecules with total mass M in the container is

$\frac{1}{2} M (v)^2_{avg} = 3/2 nRT$. (Total KE_{trans} of all molecules in container)
The average (translational) kinetic energy of a single molecule of mass m is

$\frac{1}{2} m (v)^2_{avg} = 3/2 kT$. (KE_{trans} of one molecule)
where the Boltzmann constant $k = R / N_A$.

Kinetic Theory of Gases 3

A few assumptions have been made when we constructed the theory:

1. The container has a large number of identical molecules and the walls of the container are rigid and do not move.
2. The molecules are point size particles and the volume that they occupy is negligible compared to the volume of the container.
3. Molecules are in random motion and they obey Newton's laws.
4. All collisions are perfectly elastic.
5. The molecules do not exert any force on each other except during collisions.
6. The effects of gravity can be ignored.

An Estimate

Let us now make an estimate of the average velocities of the gas molecules in LKC-LT.

Giancoli pg 480 Example 18-4 (modified)

Eight particles have the following velocities in the x-direction, given in m/s: 1.0, -6.0, 4.0, -2.0, 6.0, -3.0, 2.0, 5.0. Calculate (a) the average velocity and (b) the rms speed.

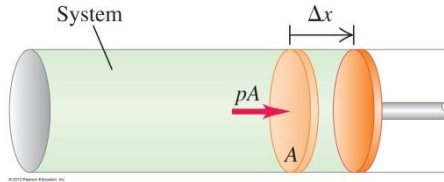
Example:

The temperature in a container of gas is at 300 K. How will the root mean-square speed of the gas change when the temperature is doubled?

At the same temperature, molecule A has a mass double of molecule B. What is the ratio of their translational kinetic energies? What is the ratio of the root mean-square speed?

Work Done by Gas

In dynamics, work done is defined as $W = F\Delta x$. We can do simple modification by noting that pressure $P = F / A$ where A is area.

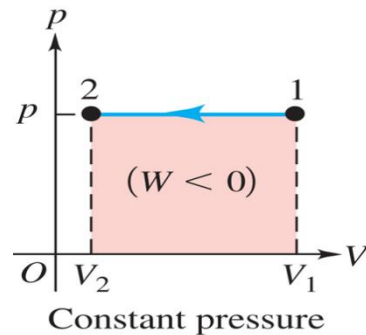
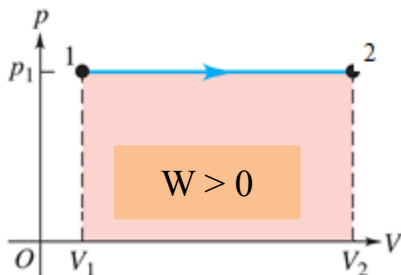


$$W = F\Delta x = PA\Delta x = P\Delta V$$

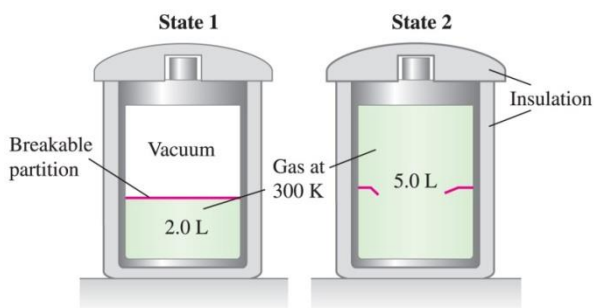
At constant pressure,

$$W = P (V_2 - V_1).$$

These processes can be represented on p-V diagrams.



When $V_2 > V_1$, work done W by gas is positive. However, if the process is a compression where $V_2 < V_1$ then work done W by gas W is negative. What about a free expansion? [The partition in state 1 breaks suddenly and all the gas fill up the container, Young (College) pg 497 Fig 15.20]

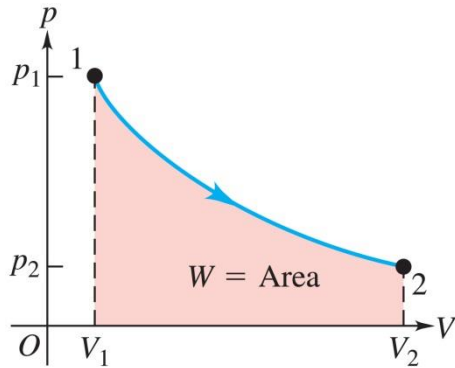


Free Expansion

1. No Temp Change
2. No Work done
3. No heat transfer (entering/leaving system)
4. No Intermediate equilibrium states

Work Done by Gas (Isothermal expansion)

If the temperature of the gas remains constant throughout an expansion, we have an isothermal expansion. The process is represented in the diagram below:



(d)

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The line joining 1 to 2 is known as an isotherm. The equation of this line is given by the ideal gas equation:

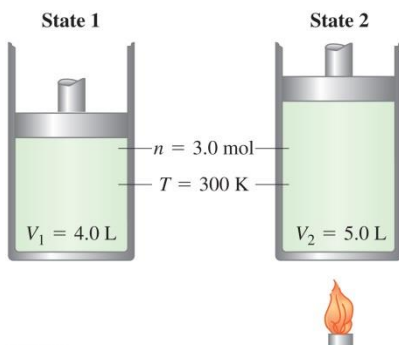
$$pV = nRT$$

The area under this line gives the work done by the gas.

$$W = \int_{V_1}^{V_2} p dV = \int_{V_1}^{V_2} \frac{nRT}{V} dV$$

$$W = nRT \ln\left(\frac{V_2}{V_1}\right)$$

Example: [Young (College) pg 500, Ex 15.9]



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The diagram shows a slow, controlled Isothermal expansion of 3.0 moles of ideal gas. The initial volume is 4.0 L and the final volume is 5.0 L. While the gas expands, heat is supplied from a flame to keep the temperature at 300K. How much work is done by the gas in the expansion? What is the change in internal energy of the gas?



Can we convert thermal energy into useful mechanical energy?

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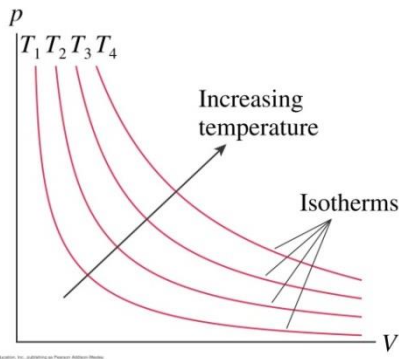
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Week 13

Serway Chap 19.4 – 19.5, 20.4, 21.1

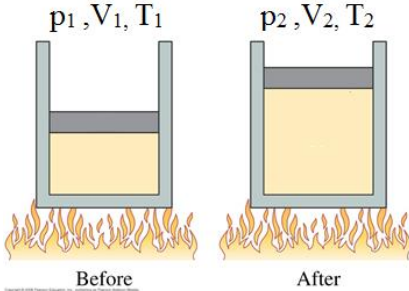
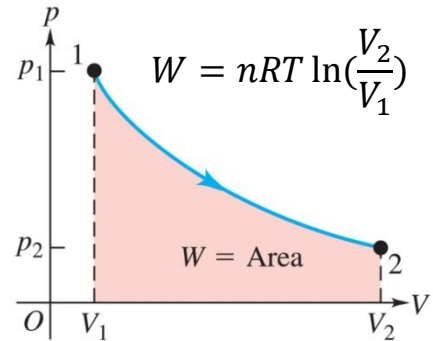
"Make everything as simple as possible, but not simpler."

- Albert Einstein (1879-1955)

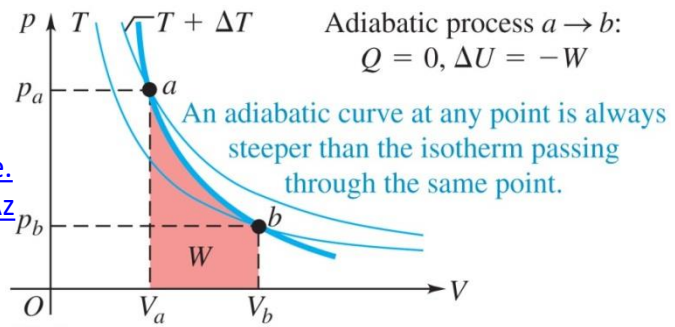


$$PV = nRT$$

$$Q = \Delta U + W$$

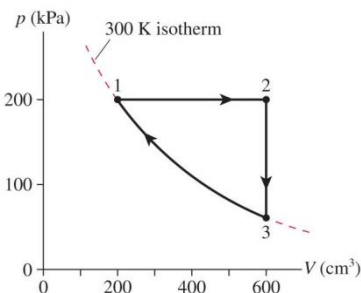


<https://www.youtube.com/watch?v=rIK7JIAz9WY&t=4s>



$$P_a V_a^\gamma = P_b V_b^\gamma = \text{constant}$$

$$T_a V_a^{\gamma-1} = T_b V_b^{\gamma-1} = \text{constant}$$



<https://www.youtube.com/watch?v=rr8g62rM7dM>

Thermal Efficiency

$$\eta = \frac{W_{out}}{Q_H} = 1 - \frac{Q_C}{Q_H}$$

Internal Energy, Heat and Work Done

In Newtonian dynamics, we have seen kinetic energy, work done and various forms of potential energies (e.g. gravitational, elastic). In thermodynamics, we have three main forms of energy to consider – internal energy U , heat Q and work W .

Internal Energy

Internal Energy (U) of a system is the sum of all the kinetic and potential energies of its constituent particles. In our ideal gas model where the gas molecules are point particles and do not interact with each other, the internal energy (U) is simply given by the kinetic energy of the gas:

Monoatomic gases, translational KE

$$U = \frac{3}{2} nRT \text{ (or } \frac{3}{2} NkT)$$

Diatomic gases (e.g. O_2 , N_2) , translational + rotational KE

$$U = \frac{3}{2} nRT + \frac{2}{2} nRT = \frac{5}{2} nRT \text{ (or } \frac{5}{2} NkT)$$

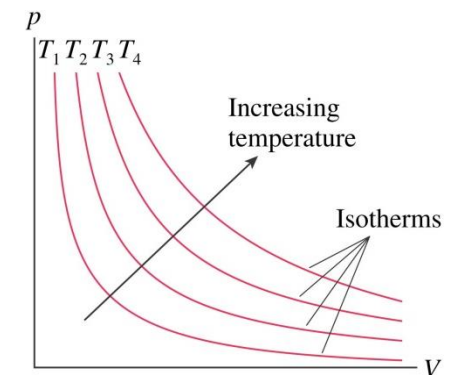


<https://www.youtube.com/watch?v=Wk9XiChXTjA&t=4s>

where n is the number of moles of gas and N is the total number of gas molecules. We note here that the internal energy is dependent on a 'state variable' temperature T . Internal energy is itself a 'state variable'.

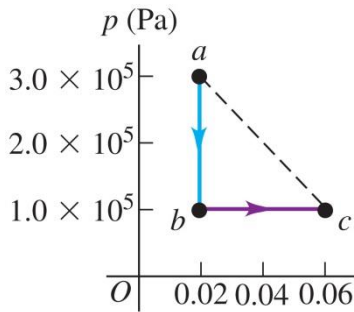
In the previous lecture, we used the p - V diagram to study work done by a gas as the volume occupied by the gas changes. We can also use p - V diagrams to study internal energies U of a system, since it is a function of temperature T . The ideal gas equation is $pV = nRT$ or $P = nRT/V$

In the p - V diagram on the right, we see isothermal lines at different constant temperatures: $T_1 < T_2 < T_3 < T_4$. Along each of these lines, the temperature is constant so the internal energy is also constant.



Taken from Knight Chap 16

Example: [Young (College), pg 512 Prob 55]



Referring to the p-V diagram on the left. Here, we have 0.005 moles of an ideal monoatomic gas.

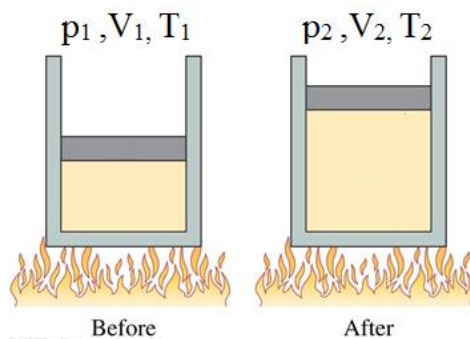
- a) Calculate the temperature at points a , b and c .
- b) Calculate the changes in internal energy for
 - i) a to b ii) b to c iii) c to a
 - and iv) $abca$

First Law of Thermodynamics

When heat is added to a system, some heat causes a rise in temperature of the system – hence there will be an increase in internal energy and some heat causes the system to expand, doing work against the surrounding. The First law of thermodynamics states that

$$Q = \Delta U + W$$

where Q is the heat supplied to a system, ΔU is the increase in internal energy and W is the work done by the system on the surrounding.



Constant volume (Isochoric) processes and molar heat capacity

We have seen in the previous lecture that work done W is given by the area under a p - V graph. i.e. $W = \int_{V_1}^{V_2} p dV$. Therefore, if the volume of a system is kept constant, no work will be done in the process. This is known as an isochoric process. (See p - V diagram below.) All the heat supplied will be used to increase the internal energy of the system.

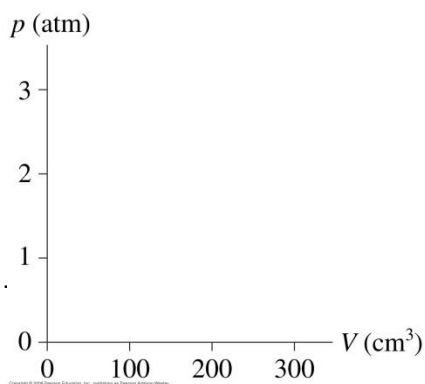
Hence, $W = 0$ and

$$\begin{aligned} Q &= \Delta U \\ Q &= \\ &= \end{aligned}$$

Here, we identify $C_v = 3R/2 = 12.47 \text{ J / (mol} \cdot \text{K)}$ as the constant volume molar heat capacity of ideal (monoatomic) gases. It is the heat required to increase the temperature of 1 mole of ideal gas by 1 K at constant volume.

Remarks:

1. The constant volume molar heat capacity C_v of ideal gases matches that of monoatomic gases such as helium and argon.
2. For diatomic gases, $C_v = 5/2 R$ is a better estimate and for polyatomic gases, C_v takes other values.



Constant pressure (Isobaric) processes

We have seen in the previous lecture that work done W is given by the area under a p - V graph. i.e. $W = \int_{V_1}^{V_2} p dV$. Thus, if the pressure is constant, it is very easy to compute the work done:

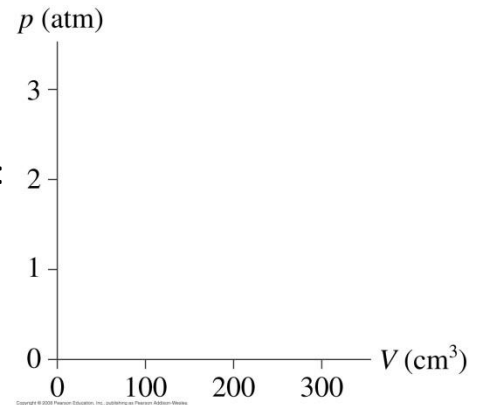
$$W = p (V_2 - V_1) = nR (T_2 - T_1).$$

It is obvious that the temperature and hence the internal energy will change from pt 1 to pt 2: (monoatomic)

$$\Delta U = 3/2 nR (T_2 - T_1).$$

Thus, in this process the heat supplied can also be easily obtained (monoatomic)

$$\begin{aligned} Q &= \Delta U + W \\ &= (3/2 + 1)nR (T_2 - T_1) \\ &= 5/2 nR \Delta T \end{aligned}$$



[Taken from Knight Chap 16]

Here, we identify $C_p = 5R/2$ as the constant pressure molar heat capacity of ideal monoatomic gases. It is the heat required to change the temperature of 1 mole of ideal gas by 1 K at constant pressure. For diatomic gases, $C_p = 7R/2$.

Mayer's relation and γ

Thus, we note the simple relation between C_p and C_v

$$C_p - C_v = R$$

known as Mayer's relation. We will define the ratio

$$\gamma = C_p / C_v \quad (= 5/3 \text{ for ideal, monatomic gases})$$

$$\gamma = 1.40 \text{ (for ideal diatomic gases)}$$

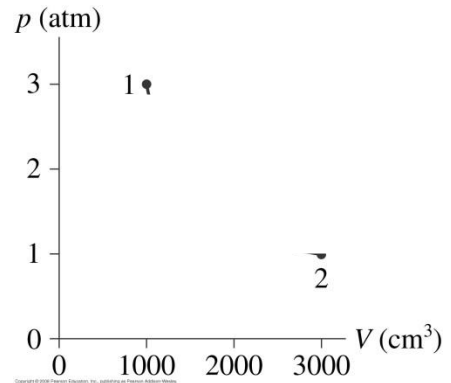
that will be used later for adiabatic process (where $Q = 0$, no heat lost to surrounding).

Constant temperature (Isothermal) processes

We have seen in the previous lecture that work done W is given by the area under a p - V graph. i.e. $W = \int_{V_1}^{V_2} p dV$. Thus, if the temperature is constant, the work done is given by

$$W = nRT \ln\left(\frac{V_2}{V_1}\right)$$

Temperature is constant from pt 1 to pt 2
for ideal gases:



Thus, in this process all the heat supplied is turned to work [Taken from Knight Chap 16]

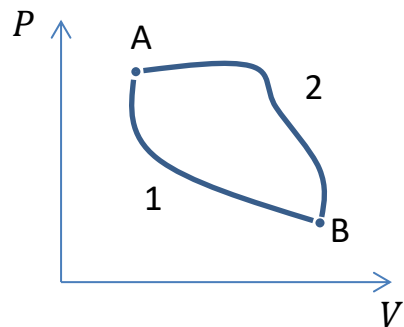
$$\begin{aligned} Q &= W \\ &= nRT \ln(V_2 / V_1) \end{aligned}$$

Remarks:

We note here that pressure p , volume V , temperature T and internal energy U are all dependent on where the state is on a p - V diagram. These quantities are known as state variables and they are related by the ideal gas equation. On the other hand, work W and heat Q are dependent on the path joining two points on a p - V diagram.

Testing your understanding

Which process has a larger change in internal energy from point A to point B, process 1 or 2?



Summary of processes

Process	Q	W	ΔU	Remarks
Isobaric				
Isovolumic (Isochoric)		0		
Isothermal			0	
	0			

A Heat Engine Example (Knight Chap 19.2)

A heat engine is a closed cycle device that extracts heat Q_H from a hot reservoir, does useful work, and then exhaust heat Q_C to the cold reservoir.

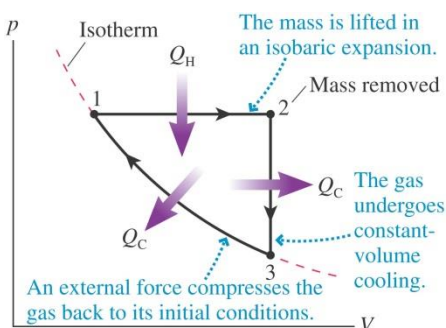


FIGURE 19.8 A simple heat engine transforms heat into work.

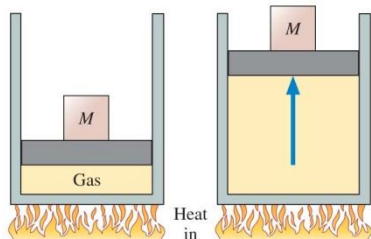
(a) Heat is transferred into the gas from the burning fuel.

(b) The gas does work by lifting the mass in an isobaric expansion.

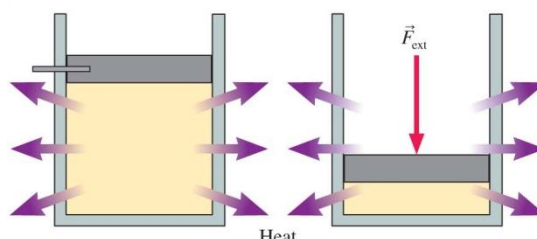
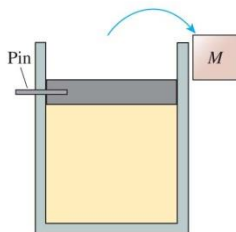
(c) The piston is locked and the mass is removed. The heat is turned off.

(d) The gas cools back to room temperature at constant volume. Then the piston is unlocked.

(e) A steadily increasing external force steadily raises the pressure in an isothermal compression until the pressure has been restored to its initial value.



Isobaric heating and expansion



Constant-volume cooling

Isothermal compression

No heat supplied/lost ($Q = 0$, Adiabatic) processes 1

When $\Delta U = 0$, we have an isothermal process; $W = 0$, we have an isochoric process. When $Q = 0$, i.e. no heat exchanges between the system and the surroundings, we have an adiabatic process. The system must be well insulated for the process to be adiabatic.

Therefore, $0 = \Delta U + W$. All work done on the gas, compression ($\Delta U = -W$) is only transformed to gain in internal energy. Alternatively, we can have $W = -\Delta U$. All work done by the gas (expansion) only results in loss in internal energy.

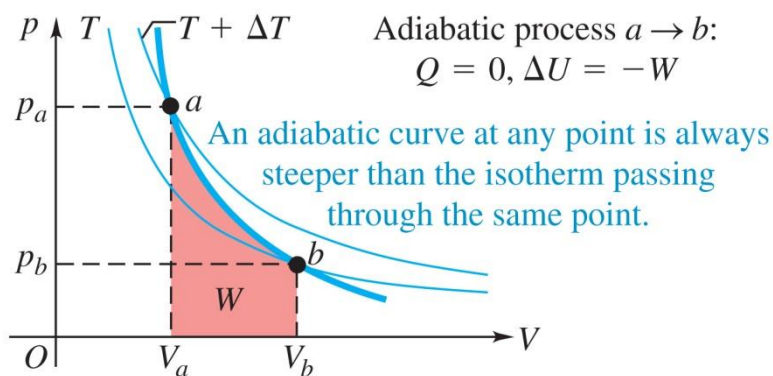
As we have seen earlier, the equation of isotherms is given by

$$PV = \text{constant}.$$

It can be shown (but we will skip the derivation) the equation of the adiabats is given by

$$PV^\gamma = \text{constant}$$

where γ is the ratio c_p / c_v . As we have seen, for monoatomic ideal gases $\gamma = 5/3 = 1.67$. Since $\gamma > 1$, we expect adiabats to be steeper than isotherms on a p-V diagram.



[Taken from Knight Chap 16]

Along an adiabat, $PV^\gamma = \text{constant}$, so referring to the diagram above;

$$P_a V_a^\gamma = P_b V_b^\gamma.$$

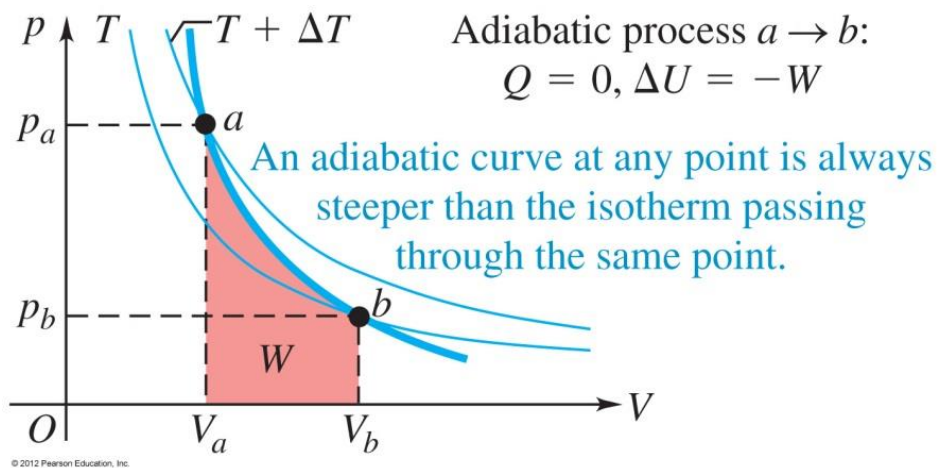
All points on the p-V diagram obey the ideal gas equation ($PV = nRT$), so we can also write

$$(nRT_a / V_a) V_a^\gamma = (nRT_b / V_b) V_b^\gamma$$

to get

$$T_a V_a^{\gamma-1} = T_b V_b^{\gamma-1} = \text{constant}.$$

Youtube: Eco Cooler: Zero Electricity Air Cooler



Adiabatic processes 2

Since $Q = 0 = \Delta U + W$, the change in internal energy can be obtained easily by
 $\Delta U = \frac{3}{2} nR\Delta T$ (monatomic) = -W.

Alternatively, we can obtain the work done by (monatomic) gas

$$W = -\frac{3}{2} nR(T_2 - T_1) = -\frac{3}{2} (P_2 V_2 - P_1 V_1)$$

For general γ , it can be shown by proper integration $\int p dV$ (which we will not go through) that the expression for work done by gas is given by

$$W = -\frac{1}{\gamma - 1} (p_f V_f - p_i V_i)$$

which agrees with the earlier expression.

$$\gamma = \frac{5}{3} \text{ monatomic}; \gamma = \frac{7}{5} \text{ diatomic}$$

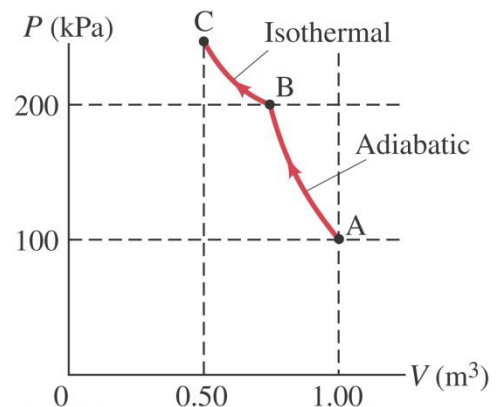
Giancoli pg 515 Example 19-12

An ideal monatomic gas is compressed starting at point A, where $P_A = 100$ kPa, $V_A = 1.00 \text{ m}^3$, and $T_A = 300 \text{ K}$. The gas is first compressed adiabatically to state B ($P_B = 200$ kPa). The gas is then further compressed from point B to point C ($V_C = 0.50 \text{ m}^3$) in an isothermal process. (a) Determine V_B . (b) Calculate the work done on the gas for the whole process.

Along the adiabatic line AB,

$$PV^\gamma = P_A V_A^\gamma = P_B V_B^\gamma \text{ where } \gamma = 5/3$$

(monatomic)



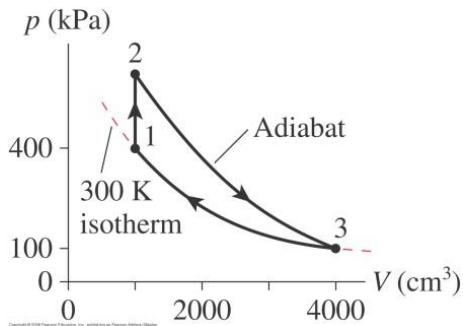
A Heat Engine Example (Knight Prob 19.56)

A heat engine uses an ideal diatomic gas that follows the pV cycle in the Figure.

- Determine the pressure, volume and temperature at point 2.
- Determine ΔU , W and Q for each of the three processes.
- How much work does this engine do per cycle and what is the thermal efficiency?

Thermal Efficiency

$$\eta = \frac{W_{out}}{Q_H} = 1 - \frac{Q_C}{Q_H}$$



Note:

i. Any point on the indicator diagram obeys $PV = nRT$.

ii. Any point on the same isotherm has the same temperature.

iii. Any point on the same adiabat obeys

$$P_1 V_1^\gamma = P_2 V_2^\gamma \text{ or } T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\text{Diatomic} \Rightarrow C_p = \quad C_V = \quad \gamma =$$

a. Number of moles of gas n : $P_1 V_1 = nRT_1$

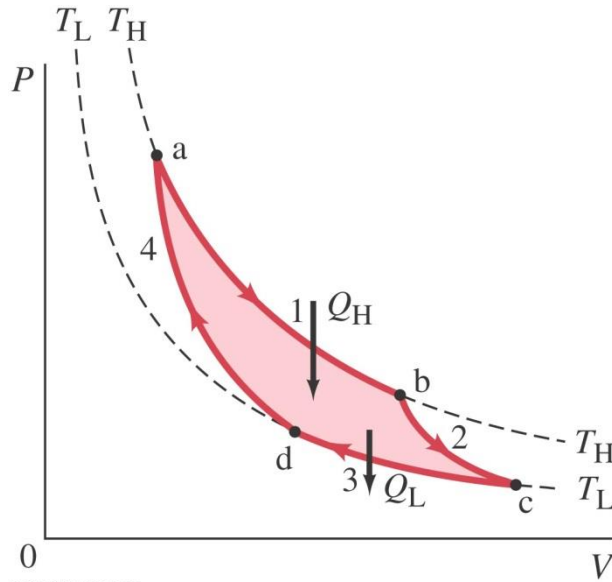
b. $W_{23} =$

$W_{31} =$

$W_{12} =$

Process				
1->2				
2->3				
3->1				
Net				

Carnot Engine (Giancoli pg 533)



The Carnot cycle / engine: (1) The gas is first expanded isothermally, with the addition of heat Q_H , along the path ab at temperature T_H . (2) Next, the gas expands adiabatically from b to c – no heat is exchanged, but the temperature drops to T_L . (3) The gas is then compressed at constant temperature T_L , path cd, and heat Q_L flows out. (4) Finally, the gas is compressed adiabatically, path da, back to its original state. No Carnot engine exists, but as a theoretical idea, it plays an important development in thermodynamics.

The Carnot cycle gives the theoretical maximum possible efficiency for real engines operating between the temperatures T_L and T_H . No engines can be more efficient than the Carnot engine.

19.56. Model: For the closed cycle of the heat engine, process $1 \rightarrow 2$ is isochoric, process $2 \rightarrow 3$ is adiabatic, and process $3 \rightarrow 1$ is isothermal. For a diatomic gas $C_V = \frac{5}{2}R$ and $\gamma = \frac{7}{5}$.

Solve: (a) From the graph $V_2 = 1000 \text{ cm}^3$.

The pressure p_2 lies on the adiabat from $2 \rightarrow 3$. We can find the pressure as follows:

$$p_2 V_2^\gamma = p_3 V_3^\gamma \Rightarrow p_2 = p_3 \left(\frac{V_3}{V_2} \right)^\gamma = (1.0 \times 10^5 \text{ Pa}) \left(\frac{4000 \text{ cm}^3}{1000 \text{ cm}^3} \right)^{7/5} = 6.964 \times 10^5 \text{ Pa} \approx 700 \text{ kPa}$$

The temperature T_2 can be obtained from the ideal-gas equation relating points 1 and 2:

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2} \Rightarrow T_2 = T_1 \frac{p_2 V_2}{p_1 V_1} = (300 \text{ K}) \left(\frac{6.964 \times 10^5 \text{ Pa}}{4.0 \times 10^5 \text{ Pa}} \right) (1) = 522.3 \text{ K} \approx 522 \text{ K}$$

(b) The number of moles of the gas is

$$R = \frac{p_1 V_1}{RT_1} = \frac{(4.0 \times 10^5 \text{ Pa})(1.0 \times 10^{-3} \text{ m}^3)}{(8.31 \text{ J/mol K})(300 \text{ K})} = 0.1604 \text{ mol}$$

For isochoric process $1 \rightarrow 2$, $W_s = 0 \text{ J}$ and

$$Q = \Delta E_{\text{th}} = n C_V \Delta T = n \left(\frac{5}{2} R \right) \Delta T = 741.1 \text{ J}$$

For adiabatic process $2 \rightarrow 3$, $Q = 0 \text{ J}$ and

$$\Delta E_{\text{th}} = n C_V \Delta T = n \left(\frac{5}{2} R \right) (T_3 - T_2) = -741.1 \text{ J}$$

Using the first law of thermodynamics, $\Delta E_{\text{th}} = \Delta W_s + Q$, which means $W_s = -\Delta E_{\text{th}} = +741.1 \text{ J}$. W_s can also be determined from

$$W_s = \frac{p_3 V_3 - p_2 V_2}{1 - \gamma} = \frac{nR(T_3 - T_2)}{1 - \gamma} = \frac{\left(\frac{4}{3} \text{ J/K} \right) (300 \text{ K} - 522.3 \text{ K})}{\left(-\frac{2}{5} \right)} = 741.1 \text{ J}$$

For isothermal process $3 \rightarrow 1$, $\Delta E_{\text{th}} = 0 \text{ J}$ and

$$W_s = nRT_1 \ln \frac{V_1}{V_3} = -554.5 \text{ J}$$

Using the first law of thermodynamics, $\Delta E_{\text{th}} = -W_s + Q$, $Q = W_s = -554.5 \text{ J}$.

	$\Delta E_{\text{th}} \text{ (J)}$	$W_s \text{ (J)}$	$Q \text{ (J)}$
$1 \rightarrow 2$	741.1	0	741.1
$2 \rightarrow 3$	-741.1	741.1	0
$3 \rightarrow 1$	0	-554.5	-554.5
Net	0	186.6	186.6

(c) The work per cycle is 186.6 J and the thermal efficiency is

$$\eta = \frac{W_s}{Q_H} = \frac{186.6 \text{ J}}{741.1 \text{ J}} = 0.25 = 25\%$$